10.3 Pharmacokinetics

The median pharmacokinetic parameters of ustekinumab following a single SC administration in adult patients with psoriasis are shown in Table 14. The pharmacokinetic parameters of ustekinumab (CL/F, V_z/F, and $t_{1/2}$) were generally comparable between 45 mg and 90 mg subcutaneous doses.

Table 14. Summary of Pharmacokinetic Parameters of Ustekinumab Following a Single 45 or 90 mg Subcutaneous Administration in Adult Patients with Psoriasis

Dose	45 mg			90 mg		
PK parameter	**n**	**Median (Range)**	**Mean (± SD)**	**n**	**Median (Range)**	**Mean (± SD)**
C_{max} (mcg/mL)	22	2.4 (1.0, 5.4)	2.7 (± 1.2)	24	5.3 (1.2, 12.3)	6.1 (± 3.6)
t_{max} (day)	22	13.5 (1.9, 58.2)	15.3 (± 13.5)	24	7.0 (2.9, 27.1)	9.9 (± 7.4)
AUC (mcg·day/mL)	18	84.9 (31.2, 1261.9)	196.7 (± 298.2)	21	226.9 (57.1, 755.5)	274.9 (± 206.5)
$t_{1/2}$ (day)	18	19.8 (5.0, 353.6)	45.6 (± 80.2)	21	21.2 (13.6, 85.8)	26.7 (± 19.3)
CL/F (mL/day/kg)	18	5.3 (0.2, 12.9)	5.8 (± 3.5)	21	4.5 (1.5, 14.9)	5.7 (± 3.6)
V_z/F (mL/kg)	18	154.2 (32.6, 280.5)	160.5 (± 64.5)	21	160.5 (37.3, 354.1)	178.7 (± 85.2)

Source data: C0379T04 CSR

Dose Linearity

The systemic exposure of ustekinumab (C_{max} and AUC) increased in a linear manner following a single subcutaneous administration at doses ranging from approximately 24 mg to 240 mg in patients with psoriasis.

Single Dose vs. Multiple Doses

Serum concentration-time profiles of ustekinumab were generally predictable after single or multiple subcutaneous dose administrations on the basis of a one-compartment model. In patients with psoriasis, steady-state serum concentrations of ustekinumab were achieved by Week 28 after initial subcutaneous doses at Weeks 0 and 4 followed by doses every 12 weeks. The median steady-state trough concentration ranged from 0.21 mcg/mL to 0.26 mcg/mL (45 mg; n = 242 to 390) and from 0.47 mcg/mL to 0.49 mcg/mL (90 mg; n = 236 to 386) in patients with psoriasis. There was no apparent accumulation in serum ustekinumab concentration over time when given subcutaneously every 12 weeks.